

An explicit modular approximation to $\zeta(5)$, with its exact asymptotics

River & Claude (Fable) — modular Apéry project, level 12 CM-isogeny system — 2 September 2026

The host. On $X_0(12)$ take the Hauptmodul $h = \eta_1^3 \eta_4 \eta_6^2 / (\eta_2^2 \eta_3 \eta_{12}^3) = q^{-1} - 3 + 2q + \dots$ and the Fricke-invariant coordinate

$$x = \frac{h}{(h+3)(h+4)} = q - 4q^2 + 2q^3 + 12q^4 - 17q^5 - \dots, \quad x(i/\sqrt{12}) = x_+ = 7 - 4\sqrt{3}.$$

The map $q \mapsto x$ folds at the Fricke point $\tau_* = i/\sqrt{12}$; the only finite singularities of everything below are $x = -1, 7 \mp 4\sqrt{3}$.

The source and its companions. The weight-6 Eisenstein combination $\tilde{F} = -\frac{1}{504}(E_6 - 176E_6(2\tau) + 2079E_6(3\tau) - 4928E_6(4\tau) + 4752E_6(6\tau) - 1728E_6(12\tau)) = q - 143q^2 + 2323q^3 - \dots$ has Dirichlet polynomial $P(s) = 1 - 176 \cdot 2^{-s} + 2079 \cdot 3^{-s} - 4928 \cdot 4^{-s} + 4752 \cdot 6^{-s} - 1728 \cdot 12^{-s}$ with $P(0) = P(2) = P(3) = P(4) = P(6) = 0$: every period except $\zeta(5)$ is annihilated, and $L(\tilde{F}, 5) = P(5)\zeta(5)\zeta(0) = \frac{11}{144}\zeta(5)$. Let $f = \sum A_n n^{-5} q^n$ be its fifth Eichler integral ($\tilde{F} = \sum A_n q^n$), let $E = \frac{1}{7}(-9E_4(3\tau) + 16E_4(4\tau))$ (anti-Fricke, weight 4) and $Q(x) = 143x^3 + 189x^2 + 21x + 7$. Define

$$U = \frac{7E}{Q(x)} = \sum_{n \geq 0} c_n x^n, \quad V = U \cdot f = \sum_{n \geq 0} d_n x^n.$$

Then $c_n \in \mathbb{Z}$ and $D_n^5 d_n \in \mathbb{Z}$ with $D_n = \text{lcm}(1, \dots, n)$:

$$c_n : 1, -3, -18, -194, -2025, -22125, -249284, \dots \quad \frac{144}{11} \frac{d_n}{c_n} : -\frac{48}{11}, \frac{111}{44}, \frac{99409}{115236}, \frac{4516373}{4276800}, \dots$$

Both sequences satisfy one explicit 14-term recurrence $\sum_{j=0}^{14} P_j(n) y_{n-j} = 0$ (resp. $= 13 \cdot 2^{18} (-1)^{n+1}$ for d_n) with $\deg P_j = 5$, $P_0(n) = n^5$, $P_{14-j}(n) = -P_j(11-n)$, e.g. $P_1(n) = -66n^5 + 305n^4 - 570n^3 + 550n^2 - 269n + 53$; its characteristic polynomial is $(\lambda+1)^4(\lambda^2 - 14\lambda + 1)^5$, i.e. the roots are -1 and $7 \pm 4\sqrt{3}$, the reciprocals of the singular values.

Theorem (limit and rate).

$$\frac{144}{11} \frac{d_n}{c_n} \rightarrow \zeta(5), \quad \frac{d_n}{c_n} - \frac{11}{144} \zeta(5) \sim \frac{13}{96K} (-1)^n (7 - 4\sqrt{3})^n n^{1/2} (\log n)^4,$$

where $c_n \sim -K(7 + 4\sqrt{3})^n n^{-3/2}$ and the constant K is, in closed form,

$$K = \frac{\Gamma(1/3)^{12}}{\pi^{17/2}} \cdot \frac{3^{8/3} (15\sqrt{3} - 23) \sqrt{(12 + 7\sqrt{3})/2}}{2^{37/3} (2205 - 1273\sqrt{3}) (45 + 26\sqrt{3})^{1/3}} = 0.68537184849053440595143004488 \dots$$

Why. Fricke invariance makes $V - \frac{11}{144}\zeta(5)U$ regular at the fold x_+ , so the linear form is driven by the next singularity $x = -1$, where the regularised solution behaves as $\frac{13}{480} \log^5(1+x)$; that gives the $(\log n)^4/n$ law for $d_n - \frac{11}{144}\zeta(5)c_n$. The denominator c_n has a square-root singularity at x_+ whose amplitude is a derivative at the CM point: $K^2 = 49x_+ (DE)(\tau_*)^2 / (2\pi Q(x_+)^2 (-D^2x)(\tau_*))$, $D = qd/dq$. Two facts make it a Chowla–Selberg value: $E(\tau_*) = 0$ (the anti-Fricke form vanishes at the Fricke point, so $DE(\tau_*)$ is a genuine weight-6 CM value), and $D^2x(\tau_*) = x''(2\sqrt{3})(Dh)(\tau_*)^2$ with Dh a genuine weight-2 form. With $\Omega^2 = \Gamma(1/3)^6/\pi^4$ one finds $DE(\tau_*)/\Omega^6 = \frac{243(23-15\sqrt{3})}{14336}$ and $(Dh(\tau_*)/\Omega^2)^3 = -\frac{81(45+26\sqrt{3})}{2048}$ (discriminant -48 : a conductor-4 point, hence the cube root); the formula follows, and agrees with the recurrence to 51 digits.

Numbers. Exact c_n, d_n to $n = 400$ from the recurrence, 540-digit arithmetic:

n	digits of $\zeta(5)$ in $\frac{144d_n}{11c_n}$	$ d_n/c_n - \frac{11}{144}\zeta(5) ^{1/n}$	error / leading asymptotic
5	2.7	0.1708	25.7
10	8.0	0.1225	11.9
20	19.0	0.0984	7.67
50	52.8	0.0834	5.37
100	109.7	0.0780	4.46
200	223.7	0.0751	3.86
400	452.2	0.0736	3.44

The n -th root of the error tends to $7 - 4\sqrt{3} = 0.07180$; the ratio to the leading term $\frac{13}{96K} (7 - 4\sqrt{3})^n \sqrt{n} (\log n)^4$ tends to 1 only logarithmically, as the $(\log n)^3, \dots$ corrections dictate.

What it does and does not do. The rate $\log \frac{1}{7-4\sqrt{3}} = 2.634$ against the denominator growth $\log D_n^5 \sim 5n$ gives no irrationality proof (Apéry's $\zeta(3)$ form wins with $4\log(1+\sqrt{2}) = 3.53 > 3$; here $2.63 < 5$). What is new is that the system is fully explicit — modular source, companion, recurrence, limit, and the complete leading asymptotics with a closed-form constant — and that every period other than $\zeta(5)$ has been removed by construction. Files: `consolidation/ZETA5_K.CLOSED.FORM.md`, `lattice/zeta5.K/`, and the level-12 research notes in `packages/`.